Exact classical sampling of IBM's 70-qubit graph-state advantage experiment, and where the construction becomes hard

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Code, samples and raw outputs: https://github.com/chayworks/exact-graphstate-sampling (Apache-2.0).

Technical note, draft of 2 November 2026, revised after an internal referee pass (`paper/reviews/`). Every number in this note is read from a committed artifact listed in the appendix, including the outputs of the test scripts (`results/tests/`), and every verification step has a negative control that fails as intended.

Abstract

IBM and the University of Chicago posted a 70-qubit, depth-70 random-graph-state sampling experiment as a candidate for verifiable quantum advantage (Quantum Advantage Tracker #151), with extrapolated classical costs of 10^{25} s (matrix product states) and 10^{7} s per sample (extended stabilizer simulation). We sample the instance exactly from its ideal Born distribution with a cut-rank dynamic program on the underlying graph state, at 13 s per sample on one 16-core CPU node and 18 s per sample on a laptop with an 8.6 GB table, and compute the exact probability of each of the 1,726 published hardware shots: linear cross-entropy 0.466 ± 0.031, a quantity the instance's documentation said could only be extrapolated. The width that governs the method grows as about n/2 – 4 for this construction, so memory doubles every two qubits; we sample a 76-qubit instance exactly and an 80-qubit instance with a capped hybrid sampler on one node, and note that any graph state measured in any product basis is samplable with 2^{⌊n/2⌋} table entries. The graph state is a [[70, 0, 15]] code, so every marginal on at most 14 qubits is exactly uniform. Writing each T gate of IBM's doped sibling instance as a projected ancilla brings it into the same framework at width 34: its exact amplitudes, computed on one node, agree with a 256-GPU tensor-network computation string by string. Concurrent work (tracker #261) samples the same 70-qubit instance exactly on 8 GPUs with 32 times the memory used here.

1. Introduction

In 2026 IBM and the University of Chicago posted a 70-qubit, depth-70 random-graph-state sampling experiment as a candidate for verifiable quantum advantage (Quantum Advantage Tracker #151), with a measured state fidelity of 0.37 and extrapolated classical costs of 10^{25} s (matrix product states) and 10^{7} s per sample (extended stabilizer simulation). We show that the instance is sampled **exactly from the ideal Born distribution** by a cut-rank dynamic program: **13.0 s per sample on one 16-core CPU node** (1,012 samples, 34 GB), and **18 s per sample on a laptop** with an 8.6 GB table. No rejection, no truncation, no GPU. Concurrently, tracker entry #261 (Qiu, 2 November 2026) reports exact samples of the same instance on 8 H200 GPUs at 2.42 s per sample by splitting the qubits into two halves of 35 and drawing one half from a 2^{35} conditional table (256 GiB): that is the ceiling of item 3 below with no order search, 32 times the memory used here, and it scores IBM's shots at the same 0.466 ± 0.031.

Our results on the laptop, at 76 and 80 qubits, and on the doped instance (item 4) are not reached by that method.

Four further results follow from having the exact probabilities in hand.

1. **IBM's own hardware shots, scored exactly.** The full-depth linear cross-entropy benchmark, which the instance README says "can only be extrapolated", is computed for every one of the 1,726 published shots: **XEB = 0.466 ± 0.031**, controls consistent with zero, and every per-shot probability agrees with an independently written program (tracker #260) to 10^{-5}. Two fidelity estimators built on the exact probabilities give 0.46 ± 0.04 and 0.48 [0.43, 0.54], above IBM's direct-fidelity estimate of 0.37 (95% lower bound 0.092) and much narrower.
2. **The frontier, measured rather than projected.** The width that governs the attack grows as 0.49·n − 2.9 for this construction at depth n, so memory doubles every two qubits. We sampled a 76-qubit, depth-76 instance exactly on the same 16-core node class (19 samples, 190 s each, 69 GB), with two samples reproduced to 10^{-7} by an independent engine. An 80-qubit, depth-80 instance (width 35, a 275 GB table) then ran on one 256 GiB node by slicing one vertex with a translation of the dual table (§3.4), using the hybrid sampler of §7 (60 bits proposed and accepted against their exact marginal, 20 drawn exactly): 79 samples at 246 s per proposal, 79 of 80 proposals accepted, every marginal within 0.3% of the uniform value against a cap of 3%. Its output is the Born law conditioned on the marginal staying under the cap; the excluded mass is small under a model but not proven, so we do not call these samples exact. Two were re-scored by a second slicing of the same engine to $5{\cdot}10^{-7}$.
3. **A ceiling that holds for every graph.** Because a cut rank never exceeds min(|P|, n − |P|), *any* n-qubit graph state measured in *any* product basis is exactly samplable with 2^{⌊n/2⌋} table entries: 275 GB at 70 qubits, 2.2 TB at 76, 8.8 TB at 80, 281 TB at 90, 9 PB at 100. So no graph-state sampling experiment with product measurements escapes a 2 TB node below 76 qubits, a 32 TB node (the largest sold today) below 84, or a 281 TB machine below 90, whatever graph is chosen. The random brickwork construction sits 3–5 below this ceiling at the search budgets we used (its fitted width stays on a 32 TB node to about 90 qubits), so the choice of graph can buy the experimenters at most a factor of about 30 in memory; the qubit count is the main lever. Random dense graphs reach the same widths as the brickwork at the same search budget (within one), so for this method the construction is as hard as a random graph.
4. **IBM's doped flagship, in the same framework.** The sibling instance with 468 T gates, whose README calls it adversarial for stabilizer and tensor-network methods alike, becomes a 538-vertex graph state in a product basis once each T gate is written as a projected ancilla, with an order of width 34 (§7). One exact amplitude is then one 128 GiB table on one node, 1,165 s with the two-table engine. Seventeen of IBM's hardware shots agree with the 256-GPU tensor-network entry #242 to a constant factor of $1.8{\cdot}10^{-4}$ (shot 1 across seven of our computations to $1.4{\cdot}10^{-6}$), #242's published amplitudes give the hardware fidelity 0.34 ± 0.03 three independent ways, above the README's 0.26 ± 0.02, and the first 13 hybrid samples of the instance were drawn on the same node (13 of 13 accepted against the cap).

Two further facts follow from the stabilizer structure. The output distribution of the instance is **exactly uniform on every set of at most 14 qubits**: the graph state is a [[70, 0, 15]] stabilizer code, equivalently a 14-uniform state, its distance 15 proven by exhaustive enumeration, so no statistic on fewer than 15 qubits can see the signal, and verification of such an experiment is global by nature. The same fact limits what low-order statistics of the hardware shots can show: a Pauli error anywhere in the circuit leaves a stabilizer state with the same group up to signs, whose marginals below weight 15 are still exactly uniform, so only readout asymmetry and non-Pauli channels can bias them. All 70 single-bit, 2,415 pair, 54,740 triple and 916,895 quadruple parities of IBM's post-selected shots are at chance (χ^2 p = 0.78, 0.36, 0.28, 0.95), which bounds such biases at about 4 percentage points per qubit or

parity (3σ with 1,726 shots) and says nothing about the Pauli-type noise that presumably dominates the fidelity loss.

The method is two standard ingredients joined by one identity: the rank-width simulability of graph-state measurements (Van den Nest et al. 2007) and perfect sampling in the style of Ferris and Vidal (2012), joined by the Schmidt decomposition of a graph state across a cut (Hein, Eisert and Briegel 2004): the post-measurement states Z^v|G⟩ are orthonormal, so every prefix marginal is a plain sum of squares, Pr(b1…b_t) = 2^{-t} Σ|P_t|². We verify the sampler against dense statevectors on small instances, against an independent amplitude algorithm on the full instance, against a closed-form stabilizer-group formula for its per-step conditionals (agreement $3{\cdot}10^{-8}$ on all 1,726 shots), and across three different dynamic-program paths.

Results at a glance

	Result	Artifact
Exact samples of IBM's 70-qubit instance	**1,012 samples, 13.0 s each**, 16 cores, peak 34.4 GB	`results/samples_nq70.txt`, `time_nq70.txt`
Same instance on a laptop	**3 samples, ~18 s each**, one 8.6 GB table	`results/laptop_samples_w30.txt`
Independent engine (different algorithm)	6 samples reproduced to 10^{-6} in $\log_2$ Pr; 10-bit-flip control moves by 3.37	`results/xcheck70_independent.txt`, `xcheck70d.txt`, `xcheck70e_laptop.txt`
IBM's 1,726 hardware shots scored exactly	**linear XEB 0.466 ± 0.031**; uniform control 0.16 ± 0.12; bit-reversed control −0.15 ± 0.10	`results/ibmxeb/analysis.txt`
Agreement with the independent code of #260	1,726 of 1,726 shots within $9.4{\cdot}10^{-6}$ in $\log_2$ Pr (median $2.8{\cdot}10^{-7}$)	`results/ibmxeb/analysis.txt`
Fidelity of IBM's run from exact probabilities	0.457 ± 0.037 (XEB) and 0.48 [0.43, 0.54] (mixture MLE); IBM's DFE 0.37, ≥ 0.092	`results/ibmxeb/analysis.txt`
76-qubit, depth-76 instance sampled exactly	**19 samples, 190 s each**, 69 GB, two reproduced to 10^{-7} by the independent engine	`results/nq76/analysis.txt`
80-qubit, depth-80 instance on one 256 GiB node (hybrid sampler, §3.4)	**79 samples, 246 s per proposal**, width 35 sliced to 34 (128 GiB) by a translation of the dual table; 79 of 80 proposals accepted, marginals 0.997–1.002 against a cap of 1.03; two re-scored by a second slicing to $5{\cdot}10^{-7}$	`results/nq80/analysis.txt`
IBM's doped flagship (468 T gates): exact amplitudes and first samples on one node	width 34 after the T-gadget; 17 shots agree with the 256-GPU tensor-network entry #242 to a constant factor of $1.8{\cdot}10^{-4}$ (shot 1 computed seven ways to $1.4{\cdot}10^{-6}$); 1,165 s per shot; 13 hybrid samples, 13 of 13 accepted	`results/doped/run11/`

Output statistics of our 1,012 samples	Porter–Thomas: KS 0.016 to $z \cdot e^{-z}$ (5% level 0.043), mean z^2 6.3 ± 0.3 against 6	`results/porter_thomas_nq70.txt`
Fine-structure check without a dynamic program	stabilizer-group formula reproduces the sampler's conditionals to $3.3 \cdot 10^{-8}$ on all shots	`results/chain/stabprefix_ibm_fwd.txt`
Ceiling for every graph	width ≤ ⌊n/2⌋: 275 GB at n = 70, 281 TB at n = 90; random graphs G(70, ½) reach width 32, brickwork 31	Proposition in §5, `results/ceiling_tightness.jsonl`
Low-order marginals of the ideal distribution	exactly uniform on every set of ≤ 14 qubits (minimum stabilizer weight 15, proven)	`results/minweight_exact.txt`, `results/minweight_k11.txt`, `results/minweight_case1_k11.txt`
Hardware shots, simulation-free test	70 single-bit, 2,415 pair, 54,740 triple and 916,895 quadruple parities at chance (χ^2 p = 0.78, 0.36, 0.28, 0.95)	`results/hw_marginals.txt`

The rest of the note is organised as follows: §1.1 states the task; §2 the method; §3 the results on the 70-, 76- and 80-qubit instances and on IBM's shots; §4 the verification; §5 the hardness frontier and the ceiling; §6 the stabilizer structure of the output and the simulation-free test of the shots; §7 the doped instance; §8 what the work does and does not show; §9 related work; and the appendix lists the artifact behind every number.

1.1 The task

Circuit `nq70_depth70_checks27_basis_fstate`, Quantum Advantage Tracker issue #151:

1. Hadamards on 70 qubits in a line.
2. Brickwork of nearest-neighbour CZ gates to CZ depth 70, with a random √X or S√X on every qubit between layers.
3. On every qubit, $T^\dagger \cdot H \cdot Rz(\theta) \cdot H$ with $\theta = \arccos(1/\sqrt{3})$, a rotation into the "face-state" basis that maximises stabilizer extent.
4. Measurement in Z.

The hardware run (IBM Boston, Heron R3, with 27 ancillas for spacetime Pauli checks) reports a post-selected fidelity of 0.37 from direct fidelity estimation of the underlying graph state, with a 95% lower bound of 0.092. 1,726 post-selected shots are published with the instance. The construction was chosen so that stabilizer-extent methods cost 10^7 s per sample and matrix-product-state methods 10^{25} s, according to the instance README.

2. Method

Graph form. The Clifford part maps to a stabilizer tableau, and then, by Gaussian elimination and local Cliffords, to $|S\rangle = \bigotimes U_j^\dagger |G_A\rangle$. Then

$\langle b|\psi\rangle = 2^{-n/2} \Sigma_x \Pi_j M_j[b_j, x_j] (-1)^{\Sigma_{i<j} A_{ij} x_i x_j}$, where $M_j = W_j U_j^\dagger$.

Cut-rank dynamic program. Fix a vertex order. After the prefix P is processed, the future depends only on $s = A[F,P] \cdot x_P \in GF(2)^{|F|}$. That vector lives in a space of dimension r = rank A[F,P], so the prefix is a table P[c] of 2^r complex numbers in a chosen basis of that space. Each step is a surjective

GF(2)-linear map (c, x_v) ↦ c′. We evaluate it in gather form: a `pdep` scatter plus a kernel of dimension at most 2. A generic matrix product state of the same cut needs bond dimension 2^r and therefore cost polynomial in 2^r; here the cost is linear in 2^r.

Exact sampling. With prefix P measured with outcomes b_P, the post-measurement state is

2^{-|P|/2} Σ_c P[c] Z^{B c} |G_F⟩.

For v ≠ 0, Z^v has no X part, so it anticommutes with some graph stabilizer and ⟨G|Z^v|G⟩ = 0. The states {Z^v|G_F⟩} are therefore orthonormal, and Pr(b_P) = 2^{-|P|} Σ_c |P[c]|². Measuring qubit by qubit in the DP order:

1. compute both candidate extensions' norms (one read-only pass);
2. draw the outcome;
3. store the chosen extension (a second pass).

This is the graph-state analogue of perfect MPS sampling (Ferris & Vidal 2012). The output of this chain-rule sampler, used for the 70- and 76-qubit instances, is the exact Born distribution up to floating-point error, and unlike rejection sampling its cost does not depend on the shape of the output distribution. Where the chain rule is unavailable (projected ancillas, §7) or too costly (a sliced table, §3.4) we use a hybrid sampler that re-introduces a capped acceptance step on a prefix marginal, and we label its output accordingly rather than as exact.

In-place layout. A second program keeps a single table and updates it in place, so the memory is 2^{width} complex64 entries rather than two tables of 2^{width+1}: 8.6 GB for the width-30 order of this instance, which is why a laptop can run it.

Cost for this instance. A greedy-plus-local-search order (`order2.py`) reaches maximum cut rank 30, the value #260 also reports; the natural order gives 32. The production sampling run used an earlier width-31 order and cost 4·Σ_k 2^{r_k} = $6.0 \cdot 10^{10}$ table updates per sample with two complex64 tables of 2^{31} entries; the later runs (laptop, hardware-shot scoring, chain mode) use the width-30 order.

3. Results

3.1 Sampling IBM's instance

1,012 samples in 13,202 s on an AWS r7a.4xlarge (16 cores, AMD EPYC 9R14), peak RSS 34.4 GB, about $4.50 of on-demand time.

- All samples are distinct.
- Linear XEB against their own exact probabilities is 1.020 ± 0.047, i.e. $2^{70} \Sigma p^2 \approx 2.02$, the Porter–Thomas value.
- The distribution of $z = 2^{70} \cdot p$ over the samples follows the Porter–Thomas size-biased law $z \cdot e^{-z}$: KS distance 0.016 (5% level 0.043), mean z^2 6.3 ± 0.3 against 6, mean $\log_2 \Pr$ −69.398 ± 0.038 against the predicted −69.390 (`porter_thomas_test.py`, `results/porter_thomas_nq70.txt`). So the ideal distribution over strings is Exp(1) as far as 1,012 samples can tell, which the fidelity estimators of §3.2 assume.

Laptop. With the in-place program, the arena is a single complex64 table of 2^30 entries (8.6 GB) for a width-30 order. On a laptop (AMD Ryzen 9 9955HX, 16 threads, container memory capped at 10 GB), 3 exact samples took 55 s including allocation, i.e. **about 18 s per exact sample** (`results/laptop_samples_w30.txt`). The in-place sampler was validated on every bitstring of small

circuits. On the full instance it reproduces the cloud samples' $\log_2 \Pr$ to $3\cdot10^{-7}$, and the independent amplitude engine confirms the first laptop sample: −68.5019327 against −68.5019326 reported (`results/xcheck70e_laptop.txt`).

3.2 IBM's hardware shots, scored exactly

The sampler's probability mode evaluates the exact Born probability of a given bitstring through the same code path as sampling. We scored all 1,726 hardware shots published with #151 on an r7a.4xlarge: 6.9 s per shot, 3 h 19 min in all, 17.2 GB peak (`results/ibmxeb/`). The run first reproduced two earlier cloud samples' $\log_2 \Pr$ to $5\cdot10^{-7}$ as a gate.

strings	**N**	**linear XEB**
IBM hardware shots	1,726	**+0.466 ± 0.031** (bootstrap 95% [+0.41, +0.53])
uniform random strings (control)	64	+0.16 ± 0.12
IBM shots with the bit order reversed (control)	64	−0.15 ± 0.10

Our $\log_2 \Pr$ agrees with #260's independently written FP64 code on every one of the 1,726 strings, to $9.4\cdot10^{-6}$ at worst (median $2.8\cdot10^{-7}$), and the XEB of +0.4660 is the same to four digits. Two programs written independently assign the same probability to each of IBM's shots. The uniform control is small (64 strings, 1.3σ from zero); it checks the sign and scale of the statistic rather than its tail, and a control as large as the shot set would cost two hours of the same node.

Normalising by the ideal value $2^{70}\Sigma p^2 - 1 = 1.020 \pm 0.047$, measured on our own exact samples, gives a fidelity estimate $F_{XEB} = 0.457 \pm 0.037$. A second estimator uses the whole distribution of $z = 2^{70}\cdot p(x)$ over the shots rather than its mean. If each shot is drawn from the ideal distribution with probability F and uniformly otherwise, the density of z over shots is $(Fz + 1 - F)e^{-z}$, and the maximum-likelihood fit gives $F_{MLE} = 0.48$ [0.43, 0.54] (profile likelihood, 95%). This model takes the ideal z over strings to be Exp(1), which §3.1 supports for this instance; the synthetic F = 1 case below (0.98 [0.95, 1.00]) shows the size of the residual model bias. The mixture fits the observed shot distribution (Kolmogorov–Smirnov distance 0.018, 5% threshold 0.033; `ibm_fidelity_mle.py`). On synthetic shots of known fidelity the two estimators return 0.00 ± 0.02 and 0.00 [0.00, 0.05] for F = 0, 0.42 ± 0.04 and 0.41 [0.35, 0.46] for F = 0.4, and 0.94 ± 0.05 and 0.98 [0.95, 1.00] for F = 1 (`fidelity_mle_synth.py`, `results/fidelity_mle_synthetic.txt`).

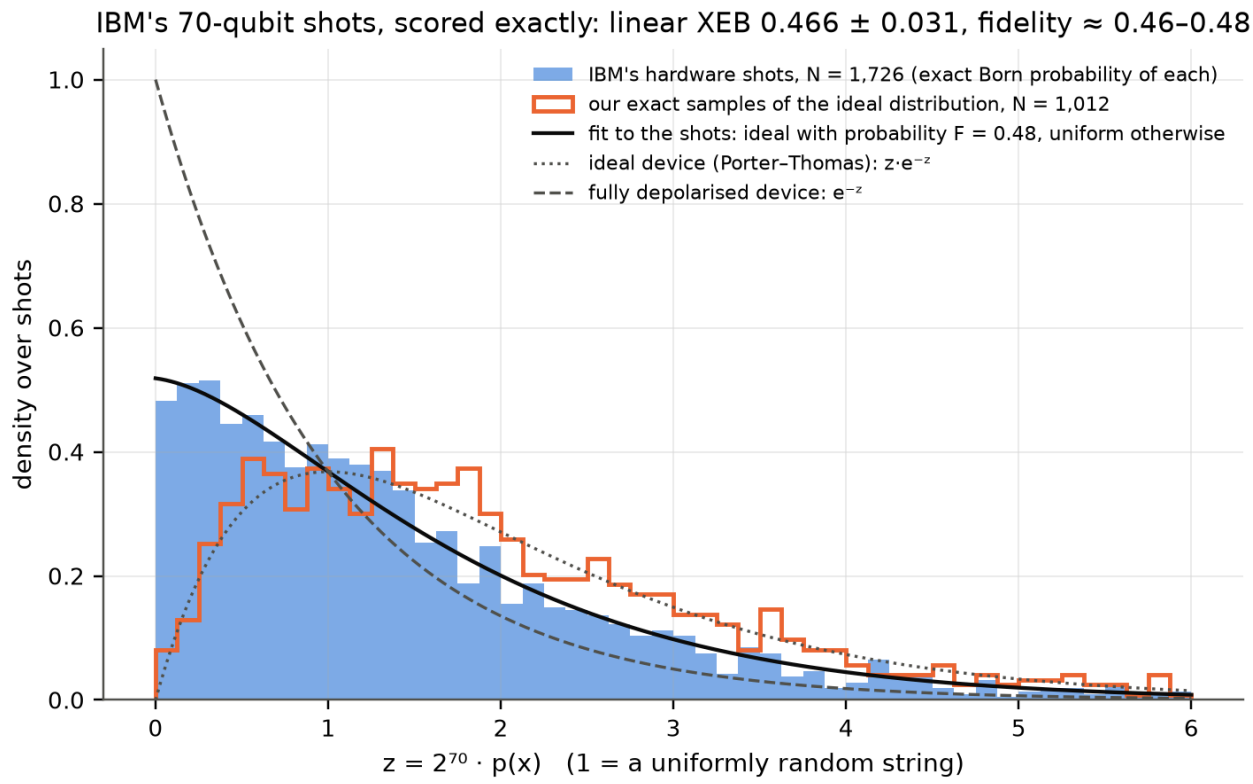


Figure 2. Distribution of $z = 2^{70}\cdot p(x)$ over IBM's 1,726 post-selected shots (exact p), our 1,012 exact samples, and the fitted clean/uniform mixture (`paper/make_figures.py`).

The instance README (tracker repository, `random_graph_sampling/README.md`) states that "XEB requires classical simulations for the outcome probabilities, hence can only be extrapolated for the full-depth circuit". Here it is computed exactly for every published shot. The shots are post-selected by spacetime Pauli checks on 27 ancillas, and the README reports a post-selected state fidelity of 0.37 by direct fidelity estimation with 80 random stabilizers, "bounded above 0.092 with 95% confidence". Our two estimates, 0.457 ± 0.037 and 0.48 [0.43, 0.54], are consistent with that bound and higher than the point estimate, with a much narrower interval. The quantities differ (the full circuit including the non-Clifford measurement layer, versus the stabilizers of the graph state; the README argues they agree to first order under Pauli twirling), and linear XEB is known to over-estimate fidelity under non-depolarising noise, so we report the gap without interpreting it. It is, in our view, the most useful thing in this note for the experimenters: an exact, per-shot benchmark of their device at full depth.

3.3 A 76-qubit instance, sampled

To test the frontier with a measurement rather than a projection, we generated a 76-qubit, CZ-depth-76 instance of the same construction (`frontier_instance.py`, seed 1076, 1,465 edges), found an order of width 33 (natural order 38), and ran the in-place sampler on the same 16-core node class (r7a.4xlarge, 128 GiB): **19 exact samples in 3,604 s, 190 s per sample, 69 GB peak** (`results/nq76/`). The node first reproduced the two 70-qubit reference probabilities as a gate. All 19 samples are distinct; their self-consistency XEB is 0.83 ± 0.22. The cross-check scheduled in that run produced nothing (its complex128 tables, 137 GB, were killed by the kernel; `results/nq76/run.log`), so it was repeated in a second run. The independent sliced amplitude engine, rebuilt with complex64 tables and itself gated on the two known 70-qubit values (−69.5085289 and −68.8761655 against −69.5085302 and −68.8761651), recomputes the first two samples' $\log_2 \Pr$ as −74.1389442 and −75.3402128 against the sampler's −74.1389441 and −75.3402127 (`results/nq76/analysis.txt`).

3.4 An 80-qubit instance, sampled on one node

The 80-qubit, depth-80 instance (seed 1080, 1,573 edges) has an order of width 35, a 275 GB in-place table that no single node in our quota holds. Slicing one vertex (fixing its variable and summing the two half-problems) lowers the width to 34, but every vertex of this dense graph (degrees 30–50) has neighbours among the 20 sampled suffix qubits, which the slicing of §7 forbids: the slice's phase on a suffix neighbour cannot be folded into that neighbour's weight, because the suffix steps run once on the summed table. The way out is a property of the table itself. The in-place table is the dual representation: its entry at logical coordinate u is the summed weight of all prefix assignments whose linear functional on the suffix is $\varphi(u) = \Sigma_i u_i B_i$, where B is the cut basis the planner stores, and the partial amplitude is $F(x_S) = \Sigma_u T(u)(-1)^{\langle \varphi(u), x_S \rangle}$. A phase $(-1)^{\langle s, x_S \rangle}$ on the suffix is therefore a translation of the table, $T(u) \to T(u + a)$ with $s = \Sigma a_i B_i$ (the shift theorem of the Walsh–Hadamard transform: multiplying by a character translates the transform). The translation exists whenever s lies in the span of B, i.e. whenever the slice vertex's row on the suffix is a combination of the remaining prefix rows; at the summation cut of a hybrid run (rank ≤ 20, spanned by 59 prefix rows) that held for every slice we planned, and the planner refuses a slice for which it does not (no fallback was needed or built). The planner writes the translation for each slice and the engine applies it when it sums the dumps (`sliced_plans.py`, `gsamp2 resume --twist`). We checked the representation entry by entry against a brute-force sum on a toy T-gadget circuit (`test_dual_table.py`, $2.4 \cdot 10^{-8}$ at worst, with the unsigned sum off by 1.41 as a control), the twisted two-slice sampler against the unsliced one on that toy ($9.1 \cdot 10^{-8}$ in $\log_2 \Pr$, with the untwisted sum off by 2.54 as a control) and at full scale on the 70-qubit instance (three proposals, identical sampled bits, $1.8 \cdot 10^{-6}$ in $\log_2 \Pr$ and $3.2 \cdot 10^{-7}$ on the prefix marginal; `test_twist.py`); the outputs of both scripts are in `results/tests/`. The same checks ran as gates on the cloud node before the 80-qubit sampling (`results/nq80/run.log`).

With vertex 34 sliced (width 34, 137 GB per slice) the sampler is the hybrid one of §7, not the chain rule: a chain-rule sample would need both slice passes at every one of the 80 steps, since the in-place table cannot be branched, about 80 times the cost of a proposal. Each proposal draws the first K = 60 bits uniformly; one pass per slice gives their exact marginal z (in units of the uniform distribution) together with exact draws of the last 20 bits, and the proposal is accepted with probability z/c, c = 1.03. One r7a.8xlarge (32 cores, 256 GiB) produced **79 samples of the 80-qubit instance at 246 s per proposal** (two passes of 2^{34} per proposal, peak resident set 128 GiB, 5.5 h of sampling; `results/nq80/`): 79 of 80 proposals accepted, z between 0.9970 and 1.0018 (the spread expected when z averages 2^{20} Porter–Thomas probabilities is 2^{-10}), none above the cap; mean $\log_2 \Pr$ −79.29 ± 0.12 against the Porter–Thomas value −79.39 and −80 for the uniform distribution; XEB against their own exact probabilities 1.02 ± 0.14, KS distance to $z \cdot e^{-z}$ 0.095 (5% level 0.153; `results/nq80/porter_thomas_nq80.txt`); all 79 strings distinct. These samples follow the Born law conditioned on z ≤ c. The excluded mass is the probability, over uniformly random 60-bit prefixes, that z exceeds 1.03: 80 proposals bound it only at the few-percent level, the observed spread of 10^{-3} puts the cap 30 standard deviations out under a Gaussian model, and we have no rigorous bound (it would need the second moment of the prefix marginal over all 2^{60} prefixes, which we did not compute), so we do not call them exact samples. The first two were re-scored with a second slicing of the same engine (vertices 10 and 40, four slice passes of 472 s), which agrees with the sampler to $5.2 \cdot 10^{-7}$ and $4.4 \cdot 10^{-7}$ in $\log_2 \Pr$; no independent engine has evaluated an 80-qubit probability (the independent engine's checks are at 70 and 76 qubits). Before sampling, the node passed two gates: the two known 70-qubit values to $2.6 \cdot 10^{-7}$, and the twisted two-slice run against the unsliced hybrid sampler on three 70-qubit proposals (identical bits, $8.2 \cdot 10^{-7}$ in $\log_2 \Pr$). A scan of 16 other circuit seeds at n = 80 found widths 35–36 only (`results/frontier_instances/fi80_cseed_scan.log`), so the slicing, not a lucky instance, is what put 80 qubits on one node.

4. Verification

1. **Small instances.** On circuits of the same form with up to 9 qubits, the distribution implied by the sampler matches the statevector on every bitstring to $2 \cdot 10^{-15}$. The C++ sampler's 20,000-shot histogram has the same total-variation distance from the exact distribution as an ideal sampler's. Every test has a negative control that fails as intended (`test_*.py`; outputs in `results/tests/`).
2. **Full instance.** A sliced amplitude engine recomputes $\log_2 \Pr$ of sampled bitstrings; it is a different algorithm from the sampler.

sample	sampler $\log_2 \Pr$	independent engine $\log_2 \Pr$
1	−69.5085302	−69.5085294
2	−68.8761651	−68.8761651
3	−68.3318209	−68.3318207
4	−69.0978566	−69.0978571
5	−70.2946875	−70.2946876

   Control: sample 1 with 10 bits flipped gives −72.8808316. That is 3.37 lower in $\log_2$ (10× less likely), against agreement within 1e-6 on the real samples.

3. **Reproducibility.** The production run first reproduced, bit for bit, three samples of a 55-qubit sub-instance (width 26) generated on a different machine, compiler and thread count (`results/check55_result.txt`).
4. **Engine-independent check.** While the cut rank of the measured prefix equals its length, the reduced state is maximally mixed, so the prefix marginal is exactly uniform; for the production order that holds for the first 30 steps. Over the 1,012 samples the first 30 step-bits show pair parities (435 tests) and triple parities (4,060 tests) at chance (4.8% and 4.4% beyond 2σ, 4.6% expected; largest $|z|$ 3.2 and 3.8). Of the 30 single-bit means, 4 fall beyond 2σ ($p \approx 0.05$), the largest at 2.4σ. A control in which one bit is replaced by a copy of another fails the test (`check_uniform_prefix.py`, `results/uniform_prefix_check.txt`). This uses only the rank profile and the sample file.
5. **Stabilizer-only conditionals.** Once the cut rank falls below the prefix length, the prefix marginal has a closed form from the stabilizer group alone: Pr(b_P) = 2^{-t} Σ_S sign(S) Π_{j∈supp S} (−1)^{b_j} n_j[S_j], summed over the 2^{t−rank} stabilizers supported on the prefix P, with n_j the Bloch vector of the measured operator. We enumerate that sum exactly while the deficit is at most 16 (steps 1–42 of the width-30 order) and compare the resulting conditionals Pr(b_k | b_1…b_{k−1}) with the sampler's chain output for all 1,726 hardware shots: the two agree to $3.3 \cdot 10^{-8}$ (half a float32 ulp at ½, the sampler's storage precision), while the conditionals themselves depart from ½ by up to $8 \cdot 10^{-5}$ at those steps (`stabprefix.py`, `results/chain/stabprefix_ibm_fwd.txt`). On small circuits the same formula reproduces the statevector marginals to $6 \cdot 10^{-15}$ on 162 prefixes, and a flipped generator sign fails it. This is a check of the dynamic program's fine structure by a method that never builds the DP table.

   The same formula explains where the information in a string lives: a stabilizer of weight w contributes at most max|n|^w, so the conditionals stay within 10^{-4} of ½ until the prefix supports many low-weight stabilizers, which in this order happens only in the last ~10 steps; the per-step spread of the chain conditionals grows by about 1.4× per step from $7 \cdot 10^{-4}$ at step 50 to 0.56 at step 70, and the last 8 steps carry 2.83 of the 2.85 variance of $\log_2 \Pr$ across the hardware shots (`results/chain/analysis_profile.txt`). For our 1,012 exact samples the picture is the same: the spread of the conditionals first exceeds 10^{-3} at step 51 (52 in the reversed order), and the last 4 and 8 steps account for 90% and 100% of the variance of $\log_2 \Pr$ (`results/chain/analysis_profile.txt`).

6. **Three dynamic-program paths agree.** Scoring every string along the width-30 order, along its reverse (the same width, a different table layout and step sequence), and, for our samples, along the width-31 production order gives the same $\log_2 \Pr$: the forward and reversed orders agree to $1.4{\cdot}10^{-5}$ at worst (median $4{\cdot}10^{-7}$) on the 1,726 hardware shots and to $3.5{\cdot}10^{-6}$ on the 1,012 samples, and the width-31 run agrees with the width-30 order to $4.3{\cdot}10^{-6}$ (`results/chain/analysis_crossorder.txt`). The stabilizer-only check of item 5 also holds along the reversed order and on the ideal samples ($3.3{\cdot}10^{-8}$, `results/chain/stabprefix_rev_ideal.txt`).
7. **Simulation-free test, and the sampler's random stream.** By §6 every marginal of the ideal distribution on at most 14 qubits is exactly uniform, so all single-bit means, pair parities and triple parities are exactly zero (for ±1 bits), and their sums of squared z-scores are χ^2 with as many degrees of freedom as tests. Our 1,012 samples: pairs $p = 0.76$, triples $p = 0.97$, quadruples $p = 0.12$, single bits $\chi^2 = 100$ on 70 ($p = 0.010$); the same run scores truly uniform random bits as a calibration and a control with one bit biased by 15% fails it ($p = 1.8{\cdot}10^{-7}$) (`hw_marginals.py`, `results/hw_marginals.txt`). We examined the single-bit excess. For the first 30 steps of the production (width-31) order the cut rank equals the prefix length, so the conditional is exactly ½ and the sampled bit must equal ($U < ½$) for the sampler's uniform draw U; reproducing the stream (std::mt19937_64 seeded with $10^6 + i$, `rngbits.cpp`) gives exactly that bit in 1,012 of 1,012 samples, and in 0 of 1,012 with the seed shifted by one (`results/rng_check.txt`). The sampler is therefore deterministic given its stream. Over those 30 exact steps the single-bit χ^2 is 36.9 on 30 ($p = 0.18$); the excess sits in steps 31–70 (63.5 on 40, $p = 0.010$). The sampled bit still coincides with the stream's sign bit through step 58 of the production order (`results/rng_check.txt`), as expected if its conditionals, like those of the width-30 order profiled in Figure 3, stay within 10^{-3} of ½ until about step 50; so a χ^2 of the stream alone over the same seeds is not independent evidence. What is independent is a direct test of the drawing step with the exact conditionals: the residuals $b_k - \Pr(b_k = 1 \mid \text{prefix})$ have mean zero and variance $p(1 - p)$ whatever the prefix, so their per-step z-scores are standard normal if each bit is drawn with its own conditional (`draw_check.py`, `results/draw_check.txt`, on the width-30 chain output of item 6). Over the 20 steps where the conditionals carry the information (51–70) this gives $\chi^2 = 23.1$ on 20 ($p = 0.29$); over steps 23–50, where the conditionals differ from ½ by less than 10^{-3}, 44.2 on 28 ($p = 0.027$); over all 70 steps 94.7 ($p = 0.026$). The test is wired to the conditionals: with the conditionals flipped ($p \to 1 - p$) on the real samples it fails at $p = 2{\cdot}10^{-4}$, a synthetic sampler drawing with p^{1.1} instead of p fails at $p = 7{\cdot}10^{-50}$, and correct synthetic draws from the same conditionals pass ($p = 0.92$). A scan of 300 blocks of 1,012 consecutive seeds shows no bias in the generator (mean χ^2 over 70 draws 70.3 ± 0.7; `results/rng_scan.txt`). The excess therefore sits in bits that are, to within 10^{-3}, the generator's own sign bits, and the drawing step passes where the dynamic program's conditionals matter. We record the single-bit $p = 0.010$ as a flag, not an explanation. A second run of 1,012 samples with each sample's generator seeded through `std::seed_seq` from (seed, i) instead of the raw integer seed + i (`gsamp2 sample2`, width-30 in-place plan, 190 min on the laptop, 11.3 s per sample; `results/samples_nq70_reseeded.txt`, `time_nq70_reseeded.txt`) shows no excess: single-bit χ^2 65 on 70 ($p = 0.65$), pairs $p = 0.31$, triples $p = 0.64$, quadruples $p = 0.64$ (`results/hw_marginals_reseeded.txt`, same script, hardware block reproduced to the digit), self-XEB 1.031 ± 0.045, Porter–Thomas KS 0.018 (`results/porter_thomas_nq70_reseeded.txt`). The $p = 0.010$ of the first run is therefore a property of that seed block, not of the drawing step, which the direct test above had already cleared.

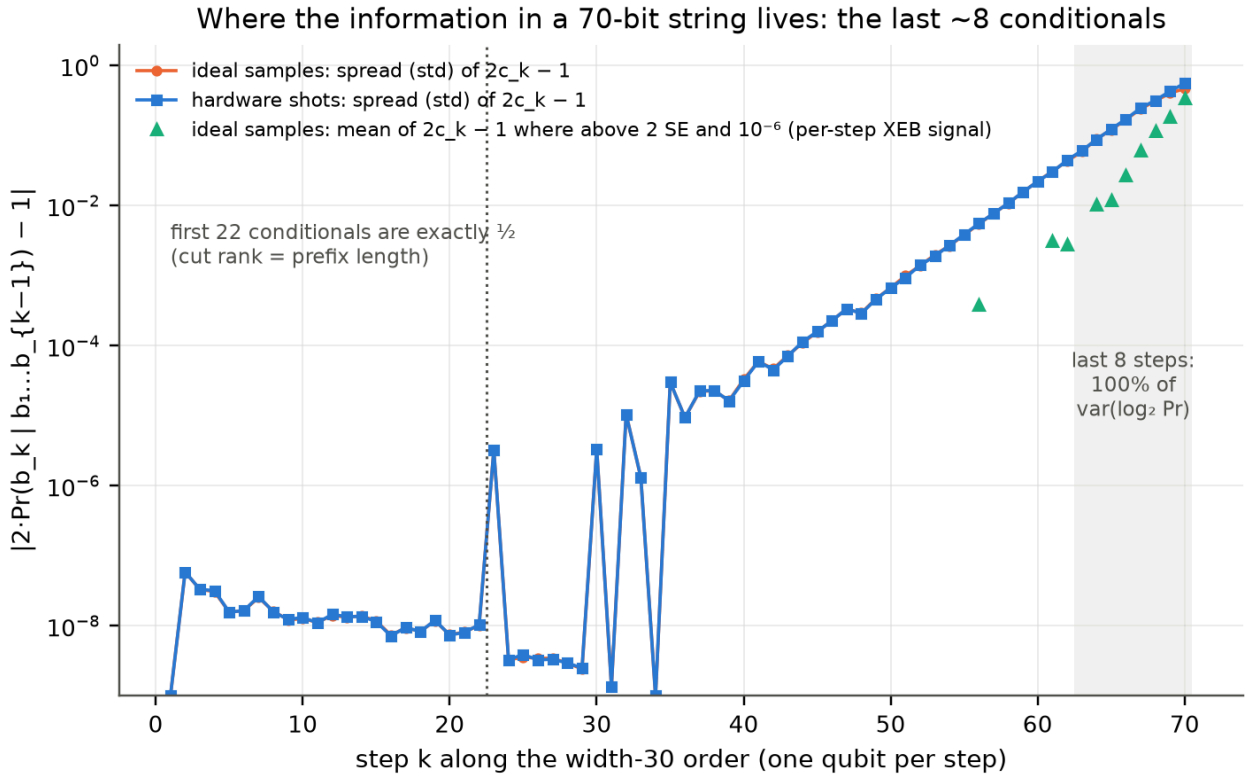


Figure 3. Spread of the exact per-step conditionals along the width-30 order for the hardware shots and the ideal samples, and the mean of 2c_k − 1 over ideal samples where it is resolved (the per-step XEB signal). The first 22 conditionals are exactly ½; the information sits in the last ~8 steps.

Per-step fidelity of the hardware shots. Dividing the hardware mean of 2c_k − 1 by the ideal mean gives a fidelity-like ratio for each informative step. Because the information sits in the last steps, only four steps per order are informative (ideal mean above 0.05): in the forward order, qubits 9, 8, 13 and 50 give 0.45 ± 0.12, 0.54 ± 0.08, 0.52 ± 0.07 and 0.48 ± 0.04; in the reversed order, qubits 45, 68, 53 and 21 give 0.47 ± 0.10, 0.43 ± 0.07, 0.43 ± 0.06 and 0.53 ± 0.05. All eight are consistent with the global fidelity estimates of 0.46–0.48 and none stands out, so the shots show no sign of a qubit that is much worse than the rest among those eight. A per-qubit map of all 70 qubits would need an order ending in each group of about four qubits, roughly 18 further width-30 passes; we did not run them.

5. Hardness frontier

We generated circuits with the #151 construction (linear chain, alternating CZ brickwork with random √X or S√X layers, CZ depth n) for n = 30–120, with one instance per n, and ran the same order search (`frontier.py`, output `results/frontier.jsonl`). At n = 70 the generated instance matches the real one: 1,192 vs 1,150 edges, natural-order width 33 vs 32, best width 31 vs 31 at the same search budget (a longer search on the real instance reaches 30).

n	best width found	memory (2 complex64 tables)	table updates per exact sample
30	12	< 1 MB	2^16.7
50	22	67 MB	2^26.4
60	26	1.1 GB	2^30.8
70	**31**	**34 GB**	**2^36.0**
76 (sampled)	**33**	**69 GB in-place**	**2^38.2**

80 (sampled, hybrid)	35, sliced to 34	275 GB in-place; 137 GB per slice	2^40.3
90	42	70 TB	2^46.3
100	47	2.3 PB	2^51.4
110	51	36 PB	2^55.9
120	56	1.2 EB	2^61.2

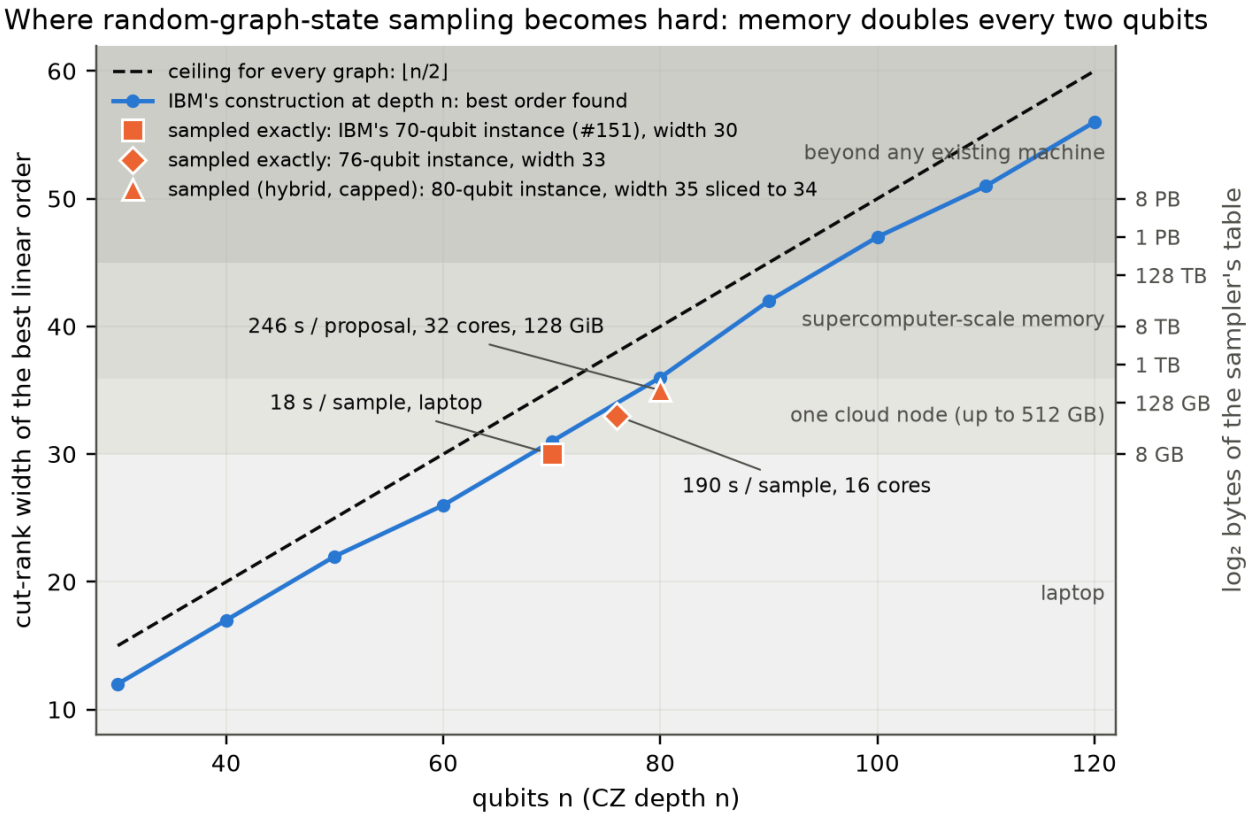


Figure 1. Best width found versus n for the construction, the ⌊n/2⌋ ceiling that holds for every graph, the three instances sampled (70 and 76 exactly, 80 with the hybrid sampler), and the memory of the in-place table. Shaded bands mark what a laptop, one cloud node of 512 GB, a supercomputer, and no existing machine can hold. The 76- and 80-qubit order searches are in `results/frontier_instances/` *(fi76_s2: width 33, log₂ Σ_k 2^{r_k} = 36.2, i.e. 2^{38.2} updates; fi80_s5: width 35, log₂ Σ_k 2^{r_k} = 38.3, i.e. 2^{40.3} updates, run at width 34 with one vertex sliced, §3.4).*

The width fits 0.492·n − 2.9; at this search budget it is between n/2 − 4 and n/2 − 3 at every n, and the longer searches on the 70-, 76- and 80-qubit instances reach n/2 − 5 (widths 30, 33, 35), so the gap to the ceiling at the budgets used is 3–5 and the true linear rank-width of these graphs is unknown. Memory doubles with every two qubits. At depth n, this method moves from one 128 GB server (n = 70) to one 256 GiB node with slicing (n = 80), to the largest single nodes sold today, 24–32 TB (n ≈ 90 at the fitted width), to a leadership-class machine (n ≈ 100: 2.3 PB with two tables), and past any existing machine beyond that.

Proposition (a ceiling for every graph). *Let |G⟩ be a graph state on n qubits and let each qubit be measured in an arbitrary single-qubit basis. For every linear order of the qubits, the cut rank of every prefix is at most ⌊n/2⌋. Consequently the sampler above draws an exact Born sample with at most 2^{⌊n/2⌋} complex table entries and at most 4·n·2^{⌊n/2⌋} table updates, whatever the graph and whatever the measurement bases.*

Proof. The cut rank of a prefix P is the GF(2) rank of the |F| × |P| adjacency block A[F,P], which is at most min(|P|, |F|) = min(t, n − t) ≤ ⌊n/2⌋. The in-place sampler's table has 2^{r_k} entries at step k and each step makes at most four passes over it. ■

The proposition is elementary, but its consequence is not widely stated: at n = 70 the ceiling is 275 GB and about $9.6 \cdot 10^{12}$ updates per sample; at the rates measured in this work ($6.0 \cdot 10^{10}$ updates in 13.0 s for the two-table sampler, 2^{38.2} updates in 190 s for the in-place sampler at 76 qubits, i.e. $4.6 \cdot 10^{9}$ and $1.7 \cdot 10^{9}$ per second on 16 cores) that is between 35 minutes and 1.6 hours per sample. The ceiling's memory is 2.2 TB at n = 76, 8.8 TB at 80, 17.6 TB at 82, 281 TB at 90 and 9 PB at 100. Against the machines that exist: a 2 TB cloud node holds the ceiling to n = 75, the largest single nodes sold today (24–32 TB) to n = 83, and a 281 TB aggregate memory, the scale of a leadership machine, to n = 90. So no graph-state sampling experiment with product measurements can escape one node below about 84 qubits, or a supercomputer below about 90, no matter how the graph is chosen. The random brickwork construction sits 3–5 below this ceiling at our search budgets (Figure 1), so better graphs cannot buy the experimenters more than a factor of about 30 in memory; the qubit count is the main lever.

Is the ceiling tight? Running the same order search (four randomised greedy orders then 3,000 local-search moves, two instances per cell; the command is recorded in `results/ceiling_tightness.cmd.txt`) on Erdős–Rényi graphs G(n, ½) and G(n, 0.3) and on the brickwork construction gives (`ceiling_tightness.py`, `results/ceiling_tightness.jsonl`):

n	ceiling ⌊n/2⌋	G(n, ½)	G(n, 0.3)	brickwork
40	20	16, 16	16, 16	16, 17
50	25	21, 21	22, 21	22, 21
60	30	26, 26	26, 26	26, 26
70	35	32, 32	32, 31	31, 31

Random graphs of either density sit 3–4 below the ceiling and the brickwork within one of them, so for this method the construction is as hard as a random graph. Whether the ceiling is tight is open: the gap of 3–5 is what this heuristic finds at this budget, and it grew by one when the search on the real instances was lengthened.

Caveats. The widths are upper bounds from heuristic orders; better orders or tree decompositions would lower them. Slicing trades memory for time; we have not measured the exchange rate on one machine (the sliced and unsliced doped runs of §7 were on different node types). The table has one instance per n. The ceiling bounds this family of methods, not all classical methods, and applies to a circuit only once it is a graph state followed by a product measurement: for the doped sibling instance that form is reached through the T-gadget of §7, on a 538-vertex graph whose ceiling (2^{269}) is vacuous; what makes it tractable there is the measured width of 34, not the ceiling.

6. Where the signal is: no marginal on fewer than 15 qubits carries any

In the graph frame every stabilizer is $S_v = X^v Z^{Av}$ with support $v \cup Av$, and by the formula of §4 item 5 the marginal of the output distribution on a qubit set Q is $\Pr(b_Q) = 2^{-|Q|} \Sigma_{S \text{ supported on } Q} \operatorname{sign}(S) \Pi_j (-1)^{b_j} n_j[S_j]$. The measured Bloch vectors have no zero component (all three components are $\pm 1/\sqrt{3}$ in the graph frame), so:

Proposition. *For the face-state basis (every Bloch component $\pm 1/\sqrt{3}$), the marginal on Q is exactly uniform if and only if no nontrivial stabilizer is supported inside Q. In particular every marginal on fewer than d_{min} qubits is exactly uniform, where d_{min} is the minimum weight of a stabilizer of the*

graph state, and a stabilizer of weight w changes a marginal by at most the fraction 3^{−w/2} of its uniform value.

Proof. "If" is the formula with the identity as the only term. For "only if", let U be a minimal nonempty support among the stabilizers inside Q and H_U the subgroup of stabilizers with support inside U; by minimality every nontrivial element of H_U has support exactly U. For any j ∈ U the map S ↦ S_j (the Pauli letter at j, modulo phase) is a homomorphism from H_U to the one-qubit Pauli group modulo phases whose kernel consists of elements with identity at j, hence of support smaller than U, hence trivial; so H_U embeds in the Klein group and has 1 or 3 nontrivial elements. The Fourier coefficient of the marginal at the character Π_{j∈U}(−1)^{b_j} is the sum over exactly those elements of sign(S) Π_j n_j[S_j] = ±3^{−|U|/2}, an odd number of terms of equal magnitude, which cannot vanish. ■ (For a general product basis with nonzero Bloch components "if" still holds and "only if" holds unless three such terms cancel exactly.)

In coding terms |G⟩ is a [[n, 0, d_min]] stabilizer code and d_min is its distance; the uniformity of every marginal on fewer than d_min qubits in every product basis is the statement that the state is (d_min − 1)-uniform, the known equivalence between k-uniform states and pure codes of distance k + 1 (Scott 2004). The "if" direction of the proposition is that fact; the "only if" for the face-state basis is the part specific to this measurement; and the minimum-weight search below is a code-distance computation.

For IBM's instance **d_min = 15 exactly**, by a three-way split that is exhaustive (`minweight_exact.py`, `results/minweight_exact.txt`): a stabilizer of weight ≤ 14 has either at most 11 X-support vertices (all $2.6{\cdot}10^{12}$ such v enumerated, lightest 15; `minweight.cpp`, `results/minweight_k11.txt`), or at most 11 Z-support vertices c = Av (all such c enumerated through a generalised inverse of the rank-68 adjacency and its 4-element kernel coset, lightest 15; `minweight_case1.cpp`, `results/minweight_case1_k11.txt`), or at least 12 of each, in which case v and Av differ in at most 4 positions and v is recovered from (A+I)v = e for the 974,120 vectors e with |e| ≤ 4 (lightest 16; A+I has full rank, so e = 0 contributes nothing and each e has one solution). The first enumeration alone cannot certify the minimum, which is why its output file ends "NOT PROVEN"; the certificate is the union of the three. The lightest stabilizer has v = {0, 5, 27, 42, 48, 49, 52, 58, 61} and support of 15 qubits. The split was validated on the 30- and 40-qubit generated instances, whose minima (6 and 8) are known from direct enumeration: it returns the same minima, and on the 30-qubit instance, run with t = 2 so that the minimiser lies outside the directly enumerated part, only case (2) finds the weight-6 stabilizer, so dropping that case would report 8 (at 40 qubits, t = 4, the direct part finds the minimiser and the other two cases return 9) (`minweight_split_test.py`, `results/minweight_split_test.txt`). So **every marginal on at most 14 qubits is exactly uniform**, and even the first non-uniform marginals deviate by at most about $3^{-7.5} \approx 3{\cdot}10^{-4}$ per stabilizer. This is the stabilizer-side statement of Figure 3: no test restricted to 14 or fewer qubits, however many shots, can tell the ideal output from coin flips; the signal lives in correlations of weight ≥ 15, and verifying such an experiment is global by nature.

The threshold grows with the size of the experiment. For generated instances of the same construction (`frontier.py` seeds, exhaustive enumeration with the proof condition met at every size; `minweight_scan.py`, `results/minweight_scan.txt`):

n	30	40	50	60	70 (IBM's instance)
minimum stabilizer weight d_min	6	8	9	12	15

So the smallest marginal that carries any signal grows roughly like n/5: at 100 qubits one would expect every marginal on about 20 qubits to be exactly uniform. The ratio d_min/n ≈ 0.19–0.21 at n = 60–70 is close to the quantum Gilbert–Varshamov rate for random stabilizer codes of rate zero (δ ≈ 0.19), as one would expect of a random-like construction; we note this as an observation.

The same fact turns the hardware shots into a simulation-free diagnostic, but of a restricted kind. A Pauli error anywhere in the circuit, propagated through the Clifford part, acts on $|G\rangle$ as a Pauli operator and leaves a stabilizer state with the same group up to signs; by the proposition its marginals on fewer than 15 qubits are still exactly uniform, whatever the error rate, and so are those of any mixture of such states. Pauli-type noise (depolarising, dephasing, Pauli errors on the gates, and the frames selected by the Pauli checks) therefore cannot bias these marginals at all. A coherent mis-rotation of the final layer cannot either, since the "if" direction holds in every product basis. What can is asymmetric readout (a bit that reads 1 more often than 0 regardless of the state) and non-Pauli channels such as amplitude damping or leakage. For IBM's 1,726 post-selected shots all 70 single-bit means (χ^2 61 on 70, $p = 0.78$), 2,415 pair parities ($p = 0.36$, largest $|z|$ 4.4), 54,740 triple parities ($p = 0.28$) and 916,895 quadruple parities ($p = 0.95$, largest $|z|$ 5.6, which is $p \approx 0.02$ after correcting for the number of tests) are at chance (`results/hw_marginals.txt`). What 1,726 shots can detect is a marginal departing from ½ by more than about 4 percentage points (3σ), so the statement is: after post-selection, readout asymmetry and other non-Pauli effects on IBM Boston are below about 4 points on every qubit and every parity of up to four qubits. The test says nothing about the Pauli-type noise that presumably accounts for most of the gap between fidelity 0.46 and 1, which is why it is consistent with, but no evidence for, global depolarisation (Figure 4).

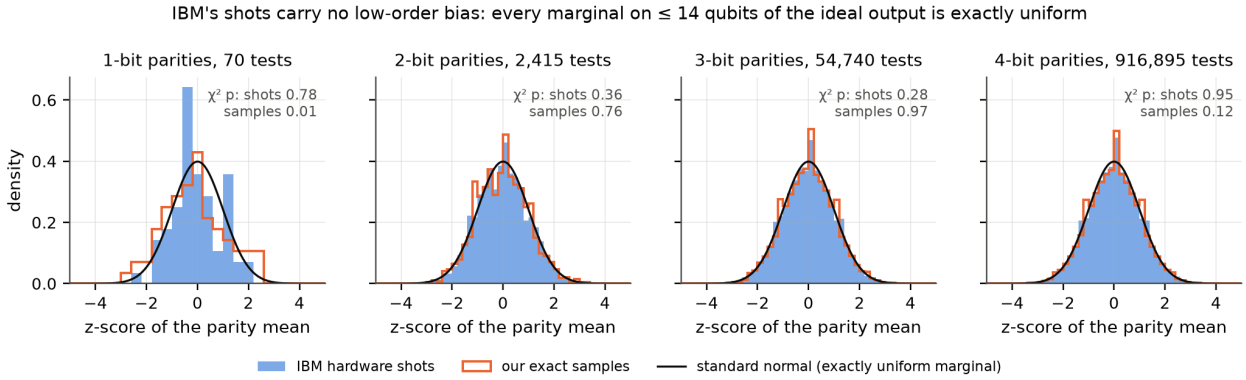


Figure 4. Distribution of z-scores of all 1-, 2-, 3- and 4-bit parity means over IBM's 1,726 shots and over our 1,012 exact samples, against the standard normal that an exactly uniform marginal implies. The χ^2 p-values are those quoted in this section for the shots and in §4 item 7 for the samples (`hw_marginals.py`). This is consistent with the mixture fit of §3.2 and is, we think, a useful fact for the experimenters.

7. The doped instance

IBM's flagship variant, `nq70_depth70_checks27_doped`, inserts 468 T gates throughout the same 70-qubit brickwork and measures in Z; its README calls it adversarial for both tensor-network and stabilizer methods (best known stabilizer rank 2^{185.5}) and reports a hardware fidelity of 0.26 ± 0.02 after post-selection. The only classical entry on the tracker computes exact amplitudes by tensor-network contraction (#242: contraction width 35, 2^{48.1} complex FMAs per batch of 256 amplitudes, all 2,051 IBM shots in 37.3 minutes on 32 nodes of eight H100 GPUs, log-XEB 0.350 ± 0.027); no entry has sampled it, and #261 states that its half/half method does not apply to it.

T gates as projected ancillas. For a data qubit q, a fresh ancilla a in $|0\rangle$, CNOT(q → a), and the projection of a onto $|\varphi\rangle = (|0\rangle + e^{-i\pi/4}|1\rangle)/\sqrt{2}$ give $\langle\varphi|_a \mathrm{CNOT}\, |\psi\rangle_q|0\rangle_a = T|\psi\rangle/\sqrt{2}$ for every $|\psi\rangle$ (the orthogonal outcome gives $ZT|\psi\rangle/\sqrt{2}$, so each projection succeeds with probability exactly ½ whatever the state). The projection is a product measurement: apply T then H to the ancilla and keep outcome 0. Hence the doped circuit is a Clifford circuit on 70 + 468 = 538 qubits followed by a product measurement, a graph state measured in a product basis, and Pr(all 468 projections succeed) =

2^{−468} exactly, so Pr_doped(b) = 2^{468}·Pr_gadget(b, anc = 0). This gadget is the T-injection that Renaud's `dcs-rankwidth` (tracker #249) applies with a linear rank-width order to a different doped instance (64 qubits, 314 T gates, width 36, probabilities on 192-core, 3 TB nodes); what is added here is its application to the 70-qubit flagship with an order of width 34 on one 128 GiB node, the string-by-string comparison with #242, and the hybrid sampler below. We checked the identity on 20 random circuits against dense statevectors (1e-10), the whole pipeline on 25 more (1e-15), and the sampler binary itself on 12 (1e-8), each with a control that drops the ancilla's T (`doped_gadget.py`, `test_doped.py`, `test_gsamp2_doped.py`).

Width. On the 538-vertex graph the natural order has width 137, but a spatial sweep (qubit 0 to 69, each qubit's ancillas in time order then its data vertex) has width **34** (`results/doped/doped_widths.json`): the cut between the left and right halves of the chain is crossed only by the CZ gates on one bond, at most 35 of them, and the sweep realises nearly that bound; simulated annealing from the sweep and single- and double-vertex slice searches found nothing lower. One exact amplitude therefore costs a table of 2^{34} complex64 (128 GiB) and 2^{40.9} table updates. For comparison, #242's contraction has width 35 and costs 2^{48.1} complex multiply-adds per batch of 256 amplitudes, 2^{40.1} per amplitude: the two operation counts per amplitude (table updates here, complex multiply-adds there; not a like-for-like FLOP count) are within a factor of two, and the gap in wall-clock time (their 37 minutes for all 2,051 shots on 256 H100 GPUs against our 26.5 minutes per shot on one 32-core node, run 11) is mostly hardware.

Why the chain rule fails here, and what replaces it. The orthonormality identity traces out the unmeasured suffix, but the ancillas must be projected, not traced (a traced ancilla dephases its T gate). Putting all ancillas before any sampled data qubit makes the prefix–suffix cut contain the whole ancilla|data cut, whose rank is 70; exact chain-rule sampling would need 2^{70} entries. Instead we move the last m = 20 data vertices to the end of the sweep order, which leaves the width at 34 and makes the final cuts 20, 20, 19, …, 1, and sample in two stages: a proposal draws the first 50 data bits uniformly; one pass gives their exact marginal Pr(prefix, anc = 0) (the 20 unmeasured data qubits are traced by orthonormality) together with exact draws of the last 20 bits from their conditionals; the proposal is accepted with probability z/c where z = 2^{50}·2^{468}·Pr(prefix, anc = 0) is the marginal in units of the uniform distribution. Because z averages 2^{20} output probabilities it is concentrated near 1 (relative spread about 2^{−10}), so with c = 1.03 almost every proposal is accepted, and the accepted output is the Born distribution conditioned on z ≤ c. The mass of prefixes with z > c is what separates this from exact sampling: z is recorded for every proposal, its observed spread puts the cap about 30 standard deviations out, and under a Gaussian model for z that mass is below 10^{-30}; but we have not bounded it rigorously (that needs the second moment of the prefix marginal over all prefixes, which we did not compute), so we call the output capped rather than exact. The difference from #260 is one of degree: there the cap sits at O(1) times the spread of the capped quantity, here at 30 standard deviations. On small T-circuits the hybrid sampler reproduces the exact output distribution (total variation at the level of an ideal sampler of the same size) and a cap set below the true maximum of z is detected as a biased output (`test_hybrid.py`). At full scale we ran it on the 70-qubit fstate instance, where the exact sampler is the ground truth: 150 proposals with the width-30 plan (50 bits proposed, 20 drawn exactly), 147 accepted, z between 0.9970 and 1.0025 (spread 0.0010, as the 2^{−10} estimate predicts), no proposal above the cap; the accepted samples' XEB against their own exact probabilities is 1.018 ± 0.128 against the exact sampler's 1.020 ± 0.047, their mean $\log_2$ Pr −69.37 ± 0.09 against −69.40 ± 0.04, and three accepted samples re-scored in prob mode agree with the sampler's reported $\log_2$ Pr to $9{\cdot}10^{-7}$ (`results/fstate_hybrid_check.txt`, `results/fstate_hybrid_samples.txt`). Each sample costs about one width-34 pass.

Full scale: the first IBM doped shot, five ways. The first of IBM's 2,051 published doped shots was scored on three rented nodes with three different arrangements of the same computation

(`graphsamp/results/doped/`, each value read from that run's output file):

run	node	plan	log₂ Pr(shot 1)	seconds per pass
7	r5a.8xlarge, 256 GiB	width 34, m = 20, unsliced	−68.295330880	9,872
7	same	width 34, m = 10 (different order)	−68.295331335	—
9	r7iz.8xlarge, huge pages	m = 20 / m = 10	−68.295331925 / −68.295330954	4,919
10	r7a.4xlarge, 128 GiB	two slices at ancillas 135, 104 (4 plans of width 33)	−68.295331107	5,888 (4 slice passes)
11 (spot)	r7iz.8xlarge, two-table engine	same two slices, two tables of 2^{33}	−68.295331101	1,592 (4 slice passes)
11 (on-demand)	r7a.8xlarge, two-table engine	same	−68.295330552	1,168 (4 slice passes)

The seven values span $1.4 \cdot 10^{-6}$ in log₂, the rounding level of complex64 tables accumulated in different orders; the two-table engine (run 11) does the same sliced evaluation in 1,592 s against 4,919 s for the in-place engine on the same machine type, 3.1 times faster per shot. Runs 9 and 10 then scored the same uniformly random control string (not a hardware shot) at −72.910036757 and −72.910036839, an agreement of $8 \cdot 10^{-8}$ between a different machine, layout and slicing; both runs ended before reaching further strings (run 9 was reclaimed by the spot market, run 10 was stopped to free its quota). An eighth value comes from a different method altogether: the tensor-network entry #242 published its amplitudes (Zenodo 10.5281/zenodo.21912448; q0-first strings, which are our Qiskit strings reversed, all 2,051 matching), and its shot 1 has log₂ p = −68.295074. The relative difference between the two methods is $1.8 \cdot 10^{-4}$ (`doped_tn242_compare.py`, `results/doped/tn242_crosscheck_run7.txt`); the 16 further shots of run 11 below show that it is a constant factor across strings. This is, to our knowledge, the first time two independent exact methods have been compared string by string on this circuit.

What the published exact probabilities say about the hardware run. Since our values agree with #242's, their 2,051 probabilities can be analysed with the estimators of §4 (`ibm_fidelity_mle.py`, `doped_tn242_analysis.py`; outputs in `results/doped/tn242_fidelity.txt` and `tn242_porter_thomas.txt`). Each of their batches also holds the 255 sibling strings that differ from the shot on qubits 0–7, so the ideal distribution itself can be tested: over 523,005 siblings z = 2^{70}p has mean 0.9991 ± 0.0014, mean z^2 2.0013 and KS distance 0.0016 to Exp(1) (5% level 0.0019), so the doped circuit's output is Porter–Thomas over strings, as the Clifford instance's is over our samples (§3.1), and E_ideal[z] = 2 is justified. For the shots: linear XEB fidelity 0.342 ± 0.028, mixture maximum likelihood 0.343 with 95% interval [0.293, 0.393], and the KS distance of the shots' z to the fitted mixture (F z + 1 − F)e^{−z} is 0.014 (5% level 0.030), so the depolarizing mixture describes the data; the log-XEB of #242 is 0.350 ± 0.027. A local check uses the conditional distribution of qubits 0–7 given the other 62 bits: the shots' mean of 256·p(x)/Σ_siblings p − 1 is 0.339 ± 0.028, the same fidelity read off eight qubits at a time. All three exceed the README's 0.26 ± 0.02, which was obtained on the device by other means, and agree with each other.

Sixteen more shots, and the first samples (run 11). Run 11 (r7a.8xlarge on-demand, 32 cores, 256 GiB; `results/doped/run11/`) first reproduced shot 1 at −68.295330552 (1,168 s, peak resident set 134,262,660 kB), then scored the last 16 of IBM's 2,051 shots at 1,165 s each with the two-table engine. Against #242 every one of the 16 differs by the same amount: log₂ difference $-2.56 \cdot 10^{-4}$ on average, all 16 within $[-2.76, -2.50] \cdot 10^{-4}$ (`tn242_compare.txt`), a relative factor of $1 - 1.8 \cdot 10^{-4}$. The offset is therefore a constant normalisation between the two pipelines, not per-string rounding; the per-string

spread of 10^{-5} is the rounding level of a 2^{48} complex64 contraction. Its origin is not identified: float32 rounding of the repeated gate constants (350 H at $1/\sqrt{2}$, 468 rz($\pi/4$)) predicts a factor of $4 \cdot 10^{-5}$ of the opposite sign, and our seven in-house values of shot 1 agree to $1.4 \cdot 10^{-6}$. It shifts #242's log-XEB by $1.8 \cdot 10^{-4}$. The 16 shots' own linear XEB is 0.33 ± 0.25 (bootstrap 95% [−0.14, 0.83]), consistent with the 0.342 ± 0.028 from all 2,051 published amplitudes (`analysis.txt`). The run then drew the first samples of the doped instance with the hybrid sampler: 13 proposals in 15,183 s, 13 accepted, prefix marginals z between 0.9981 and 1.0007 against the cap of 1.03 (mean 0.9996, sd 0.0008), 1,166 s per proposal (`samples.txt`); their $\log_2$ Pr averages −69.59 ± 0.36 (Porter–Thomas −69.39) and their self-consistency XEB is 0.81 ± 0.36, consistent with 1 at this sample size (`porter_thomas_samples.txt`). As at 80 qubits these are the Born law conditioned on z ≤ 1.03, not exact samples. A dedicated run on the same node class (run 13, about 130 proposals over 44 hours, in progress at the time of writing) extends this set; to our knowledge no other entry has sampled this instance.

8. What this does and does not show

It shows:

- the 70-qubit instance is sampled exactly, from the Born distribution, on a laptop, so it does not demonstrate a task beyond classical reach;
- the published hardware data can be benchmarked exactly at full depth, which the experiment's own README describes as impossible, and the resulting fidelity estimate is tighter than the one obtained on the device;
- the same construction stays within one node's reach to about 80 qubits (exact samples at 76; hybrid samples at 80 on a 256 GiB node, and a 32 TB node would take the fitted width to about 90), within a supercomputer's to about 90 at the ceiling, and that no choice of graph changes this by more than a few qubits (random graphs measured within one of the brickwork);
- the output distribution is exactly uniform on every set of at most 14 qubits (the graph state is a [[70, 0, 15]] code), and IBM's shots show no readout-type bias above the 4-point level that 1,726 shots can detect (Pauli noise cannot show in these marginals, so this is not a statement about the noise that sets the fidelity);
- the doped flagship's amplitudes are computed exactly on one node after the T-gadget (width 34), agreeing with the tensor-network entry #242 string by string.

It does not show:

- a new complexity-theoretic result; the rank-width reduction is from 2007 and the ceiling is elementary;
- priority on attacking this instance; #260 applied the same reduction with capped rejection sampling the day before our first run, and we cite it throughout;
- exact samples of the doped instance (its hybrid sampler is capped, as at 80 qubits), or anything about graph-state experiments with entangled measurements.

What we would suggest the experimenters run next. At depth n with n ≥ 100, the ceiling alone puts any graph beyond rank-width methods (9 PB), with no dependence on the particular graph. Doping with a few hundred non-Clifford gates, as in the sibling instance, does not: the T-gadget keeps the width at 34 there (§7), and how the width grows with the number and placement of T gates is measured at that one point only. The exact scorer here computes full-depth XEB for any graph-state run up to about 80 qubits on one node (§3.4), so an experiment at that scale can report an exact benchmark instead of an extrapolated one.

9. Related work and references

The method is the linear-order case of the rank-width simulability of graph-state measurements [1], with the Schmidt structure of graph states across a cut [2] supplying the prefix marginals and chain-rule sampling in the manner of perfect sampling for tensor networks [4]. The stabilizer-code framing of §6 is the equivalence between k-uniform states and pure codes of distance k + 1 [3]. Rank-width contraction of ZX diagrams [5] is the general form of the same idea. On the tracker, Renaud [6] applied the T-gadget with a linear rank-width order to a different doped instance (64 qubits, 314 T gates, width 36, probabilities only); Qiu [7, 8] attacked the 70-qubit instance of §1.1 with capped rejection sampling and then with exact half/half sampling on 8 H200 GPUs (1,726 samples in 1,265 s; the half/half split is the ceiling of §5 with no order search, 32 times the memory used here, and its issue states that it does not apply to the doped instances); Manabe, Gu and Pan [9] computed exact amplitudes of the doped instance by tensor-network contraction on 256 H100 GPUs and published them [10], which §7 compares string by string. The experiment itself is described in [11] and [12]; the random-graph-state hardness argument behind the construction is [13]. Our first exact samples of the 70-qubit instance are dated 29 October 2026 (`results/run.log`); [8] was posted first.

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2. M. Hein, J. Eisert and H. J. Briegel, Multiparty entanglement in graph states, Phys. Rev. A 69, 062311 (2004).
3. A. J. Scott, Multipartite entanglement, quantum-error-correcting codes, and entangling power of quantum evolutions, Phys. Rev. A 69, 052330 (2004).
4. A. J. Ferris and G. Vidal, Perfect sampling with unitary tensor networks, Phys. Rev. B 85, 165146 (2012).
5. F. Kuyanov and A. Kissinger, Efficient classical simulation of low-rank-width quantum circuits using ZX-calculus, arXiv:2603.06764 (2026).
6. K. Renaud, injected_graph_scan_nq64_depth73 (code `dcs-rankwidth`), Quantum Advantage Tracker issue #249, https://github.com/quantum-advantage-tracker/quantum-advantage-tracker.github.io/issues/249 (22 September 2026).
7. Z. Qiu, capped rejection sampling of random_graph_sampling_nq70_depth70_checks27_basis_fstate, Quantum Advantage Tracker issue #260 (30 October 2026; closed by the author on 2 November 2026 in favour of [8]).
8. Z. Qiu, Exact sampling of random_graph_sampling_nq70_depth70_checks27_basis_fstate: 1,726 samples in 1,265 s on 8 H200 GPUs, Quantum Advantage Tracker issue #261, https://github.com/quantum-advantage-tracker/quantum-advantage-tracker.github.io/issues/261 (2 November 2026).
9. H. Manabe, H. Gu and F. Pan, Classical simulation and design frontiers for IBM's doped Clifford sampling experiment, arXiv:2608.13110 (2026); tracker issue #242.
10. H. Manabe, H. Gu and F. Pan, Data for exact tensor-network simulation of IBM's doped-Clifford circuit, Zenodo, doi:10.5281/zenodo.21912448 (2026), CC BY 4.0.
11. Quantum Advantage Tracker, random_graph_sampling_nq70_depth70_checks27_basis_fstate (issue #151, 29 April 2026) and doped_random_graph_sampling_nq70_depth70_checks27 (issue #228, 9 July 2026), https://github.com/quantum-advantage-tracker/quantum-advantage-tracker.github.io.
12. S. Martiel, J.-U. Chung, A. Seif, S. Ghosh, I. Hincks, A. Deshpande, H. Manabe, H. Gu, F. Pan, B. Fefferman, J. M. Gambetta and A. Javadi-Abhari, Sampling hard circuits with verifiably high fidelity, arXiv:2607.25941 (2026).

13. S. Ghosh, D. Hangleiter and J. Helsen, Random regular graph states are complex at almost any depth, arXiv:2412.07058 (2024).

Appendix: artifacts

All paths are relative to the repository root.

- `results/samples_nq70.txt`, `run.log`, `time_nq70.txt`, `inputs.sha256`, `check55_result.txt`: the production sampling run and its reproduction check.
- `results/xcheck70_independent.txt`, `xcheck70d.txt`, `xcheck70e_laptop.txt`: independent recomputation of five cloud samples, the 10-bit control, and one laptop sample.
- `results/laptop_samples_w30.txt`: the laptop run (3 samples, timing).
- `results/ibmxeb/`: exact $\log_2$ Pr of the 1,726 hardware shots and the controls, timing, gate, input hashes, and `analysis.txt` (XEB, comparison with #260, fidelity estimators).
- `results/chain/`: per-step conditionals of all shots and samples along two orders, the stabilizer-only comparison, the cross-order comparison and the information profile.
- `results/nq76/`: the 76-qubit samples, timing, gate, independent check and `analysis.txt` (`xcheck_out.txt` holds, for the two strings listed in `xcheck_make_run6.txt`, the real and imaginary parts of the amplitude scaled by 2^{n/2}; `report_run.py` converts them as $\log_2 \Pr = \log_2(re^2 + im^2) - n$ to the values quoted in `analysis.txt`).
- `results/nq80/`: the 80-qubit hybrid-sampler run (gates, proposals with z, re-scored strings, timing, `analysis.txt`, `porter_thomas_nq80.txt`).
- `results/doped/`: the doped instance (widths, runs 7–11, the #242 comparison and the analysis of its amplitudes).
- `results/porter_thomas_nq70.txt`: the Porter–Thomas test of the 1,012 samples.
- `results/tests/`: the printed output of every `test_*.py` script and of the self-test of `doped_gadget.py`, run on the laptop on 2 November 2026 (each file ends with the script's exit code; `doped_gadget.txt` ends in a traceback because the script expects a circuit to convert after its self-test, which had already passed).
- `results/uniform_prefix_check.txt`: the engine-independent uniformity check.
- `results/minweight.txt`, `results/minweight_k11.txt`, `results/minweight_exact.txt`, `results/nq70_graph_adjacency.txt`, `results/minweight_scan.txt`: the stabilizer minimum-weight searches (IBM's instance and the generated ones) and the graph they ran on.
- `results/hw_marginals.txt`, `results/rng_check.txt`, `results/rng_scan.txt`, `results/rngbits_seeds_*.txt`, `results/draw_check.txt`: the low-order marginal test of the shots and samples, the random-stream identity check and scan, and the drawing-step test.
- `results/minweight_case1_k11.txt`, `results/minweight_split_test.txt`, `results/nq70_graph_geninverse.txt`, `results/nq70_graph_kernel.txt`: the second enumeration of the minimum-weight proof, its validation on smaller instances, and the generalised inverse and kernel it used.
- `results/fidelity_mle_synthetic.txt`: the fidelity estimators on synthetic shots of known fidelity.
- `results/ceiling_tightness.jsonl`, `results/ceiling_tightness.cmd.txt`: widths of random graphs and brickwork instances against the ceiling, and the command that produced them.
- `results/frontier_instances/`: order searches for the 74-, 76-, 78- and 80-qubit generated instances.
- `results/frontier.jsonl`: the frontier search output.
- `README.md`: the test table, the commands that reproduce each number, and the modes of the samplers.